

Lecture 3 Simple Linear Regression

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Materials in this lecture notes come from two lecture notes written by Jean Kyung Kim (Inha Univ.) and Pratheepa Jeganathan (Stanford Univ.).

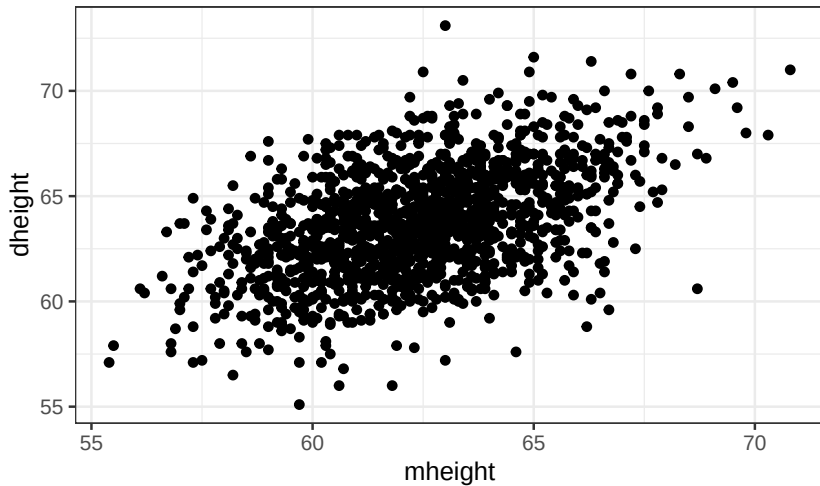


Outline

- ▶ Correlation
- ▶ Introduction to simple linear regression
- ▶ Loss functions
- ▶ Estimation
- ▶ Example



Scatter plot



Covariance and Correlation coefficient

Covariance

- ▶ Consider random pairs (X, Y) . The strength of the relationship or association between X and Y is of our main interest.
- ▶ If X and Y are continuous, the direction of the linear relationship between X and Y can be measured by **covariance**.

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$



- ▶ Given a random sample $(X_1, Y_1), \dots, (X_n, Y_n)$, the estimator of $\text{Cov}(X, Y)$ is a sample covariance.

$$\widehat{\text{Cov}}(X, Y) = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{n - 1} = \frac{1}{n - 1} S_{xy}$$

-> Sxy
->

Correlation coefficient

- ▶ If X and Y are continuous, from random sample $(X_1, Y_1), \dots, (X_n, Y_n)$ we can use Pearson correlation coefficient or nonparametric Kendall or Spearman statistics to measure the direction and strength of a linear relationship.
- ▶ Let X and Y be continuous random variables with mean μ_X, μ_Y and standard deviation σ_X, σ_Y .
- ▶ Correlation coefficient is

$$\rho = \frac{\mathbb{E}(X - \mu_X)(Y - \mu_Y)}{\sigma_X \sigma_Y} = \frac{\mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)}{\sigma_X \sigma_Y}.$$

- ▶ If X and Y are independent, $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$. Thus, $\rho = 0$, converse is not true.
 - ▶ If X and Y are bivariate normal, converse is also true.
- ▶ If X and Y are dependent, $\rho \neq 0$.
- ▶ Pearson correlation coefficient measures the linear association between X and Y .

Pearson's correlation coefficient

- ▶ Sample Pearson's correlation coefficient:

$$\widehat{Cor}(X, Y) = \hat{\rho} = r = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2 \sum_{i=1}^n (Y_i - \bar{Y})^2}} = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} = \frac{S_{xy}/(n-1)}{s_x s_y},$$

where $s_x = \sqrt{S_{xx}/(n-1)}$

- ▶ With $z_x = (x - \bar{x})/s_x$ and $z_y = (y - \bar{y})/s_y$,

$$COR(X, Y) = COV(Z_x, Z_y) = CORR(Z_x, Z_y)$$

- ▶ Compute the Pearson's correlations coefficient of heights data

```
cor(D, M, method = "pearson")
```

```
## [1] 0.4907094
```

- ▶ Examine the scatter plot.
- ▶ Interpret $\hat{\rho}$.
- ▶ $\hat{\rho}$ cannot be used for prediction purposes.

Introduction to Simple Linear Regression

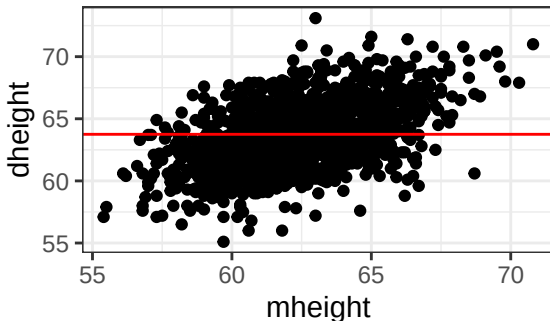


Regression

- Ignore mother's height and guessing the daughter's height, we would guess the average height of daughters

```
mean(D)
```

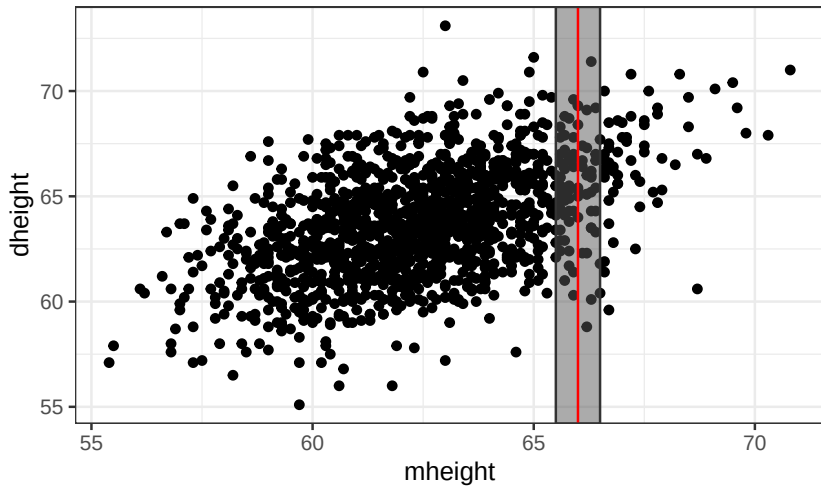
```
## [1] 63.75105
```



- Can we do better?

- ▶ A simple linear regression model fits a line through the above scatter plot in a particular way.
- ▶ Specifically, it tries to estimate the height of a new daughter in this population, say D_{new} , whose mother had height M_{new} .
- ▶ It does this by considering each slice of the data.
- ▶ Here is a slice of the data near $M = 66$, the slice is taken over a window of size 1 inch.

?(,)



```
selected_points = (M <= X+.5) & (M >= X-.5)
mean_within_slice = mean(D[selected_points])
mean_within_slice
```

```
## [1] 65.17333
```

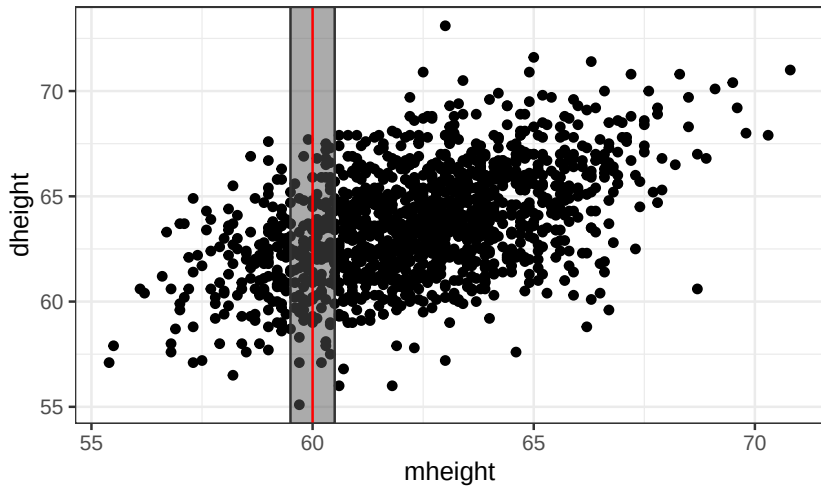
- ▶ We see that, in our sample, the average height of daughters whose height fell within our slice is about 65.2 inches.



- Of course this height varies by slice. For instance, at 60 inches:

```
X = 60
selected_points = (M <= X+.5) & (M >= X-.5)
mean_within_slice = mean(D[selected_points])
mean_within_slice
```

```
## [1] 62.42829
```



- ▶ The regression model puts a line through this scatter plot in an *optimal* fashion.
- ▶ To do this, simple linear regression assumes that the mean in slice M lies on some line

$$\beta_0 + \beta_1 M.$$

? -> (), . -> ()

- ▶ It then chooses (β_0, β_1) based on the data.


```
parameters.est = lm(D ~ M)$coef  
print(parameters.est)
```

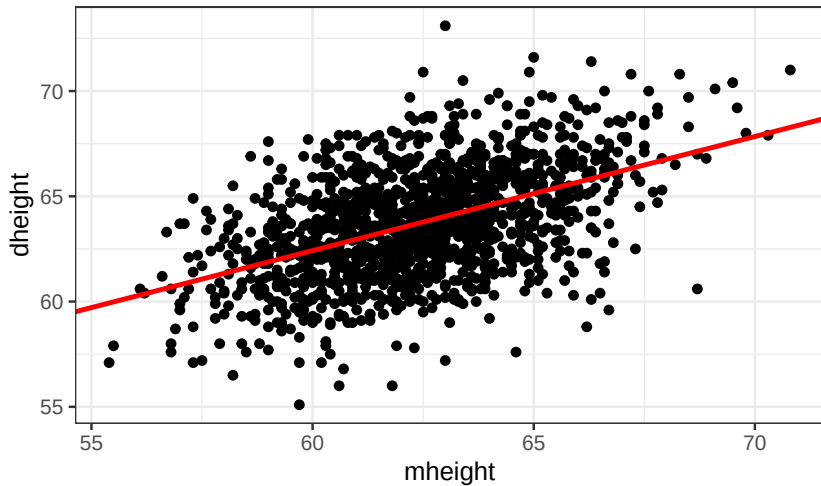
```
## (Intercept)          M  
## 29.917437      0.541747
```

```
intercept = parameters.est[1]; intercept
```

```
## (Intercept)  
## 29.91744
```

```
slope = parameters.est[2]; slope
```

```
##          M  
## 0.541747
```



Mathematical formulation

For height of couples data: a mathematical model:

$$\boxed{\text{Daughter}} = f(\boxed{\text{Mother}}) + \varepsilon,$$

where f gives the average height of the daughter of a mother of height Mother and ε is the random variation within the slice.



Linear regression models

- ▶ A *linear* regression model says that the function f is a sum (linear combination) of functions of `Mother`.
- ▶ Simple linear regression model:

$$f(\text{Mother}) = \beta_0 + \beta_1 \cdot \text{Mother}$$

for some unknown parameter vector (β_0, β_1) .

- ▶ Could also be a sum (linear combination) of *fixed* functions of `Mother`:

$$f(\text{Mother}) = \beta_0 + \beta_1 \cdot \text{Mother} + \beta_2 \cdot \text{Mother}^2$$

Simple linear regression model

- ▶ Let Y_i be the height of the i -th daughter in the sample, X_i be the height of the i -th mother.
- ▶ We have a sample of $(X_1, Y_1), \dots, (X_n, Y_n)$.

- ▶ Model:

$$Y_i = \underbrace{\beta_0 + \beta_1 X_i}_{\text{regression equation}} + \underbrace{\varepsilon_i}_{\text{error}},$$

where ε_i are random error.

- ▶ $\mathbb{E}[\varepsilon_i] = 0$ and $\mathbb{V}[\varepsilon_i] = \sigma^2$
- ▶ iid (independently, identically distributed)
 $\varepsilon_i \sim N(0, \sigma^2)$ specifies a *distribution* for the Y 's given the X 's.
- ▶ i.e. $Y_i | x_i \sim N(\beta_0 + \beta_1 X_i, \sigma^2)$ is a *statistical model*..

Fitting the model

- ▶ We will be using *least squares* regression.
 - ▶ This measures the *goodness of fit* of a line by the sum of squared errors, SSE.
- ▶ Least squares regression chooses the line that minimizes

$$\text{SSE}(\beta_0, \beta_1) = \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 \cdot X_i)^2.$$

- ▶ In principle, we might measure goodness of fit differently by sum of absolute deviation (SAD):

$$\text{SAD}(\beta_0, \beta_1) = \sum_{i=1}^n |Y_i - \beta_0 - \beta_1 \cdot X_i|.$$

- ▶ For some *loss function* L we might try to minimize

$$L(\beta_0, \beta_1) = \sum_{i=1}^n L(Y_i - \beta_0 - \beta_1 X_i).$$

Loss Functions



Why least squares?

- ▶ With least squares, the minimizers have explicit formula
 - ▶ not so important with today's computer power – especially when L is convex.
- ▶ Resulting formula are *linear* in the outcome Y . This is important for inferential reasons.
 - ▶ For only predictive power, this is also not so important.
- ▶ If assumptions are correct, then this is *maximum likelihood estimation*.
- ▶ Statistical theory tells us the *maximum likelihood estimators* (MLEs) are generally good estimators (consistency, asymptotic normality).

Choice of loss function

- ▶ The choice of the function we use to measure goodness of fit, or the *loss* function, has an outcome on what sort of estimates we get out of our procedure.
- ▶ For instance, if, instead of fitting a line to a scatter plot, we were estimating a *center* of a distribution, which we denote by μ , then we might consider minimizing several loss functions.

- ▶ If we choose the sum of squared errors:

$$SSE(\mu) = \sum_{i=1}^n (Y_i - \mu)^2.$$

- ▶ Then, we know that the minimizer of $SSE(\mu)$ is the sample mean of Y .
- ▶ On the other hand, if we choose the sum of the absolute errors

$$SAD(\mu) = \sum_{i=1}^n |Y_i - \mu|.$$

- ▶ Then, the resulting minimizer is the sample median of Y .

- ▶ Both of these minimization problems also have *population* versions as well.
- ▶ For instance, the population mean minimizes, as a function of μ

$$\mathbb{E}((Y - \mu)^2)$$

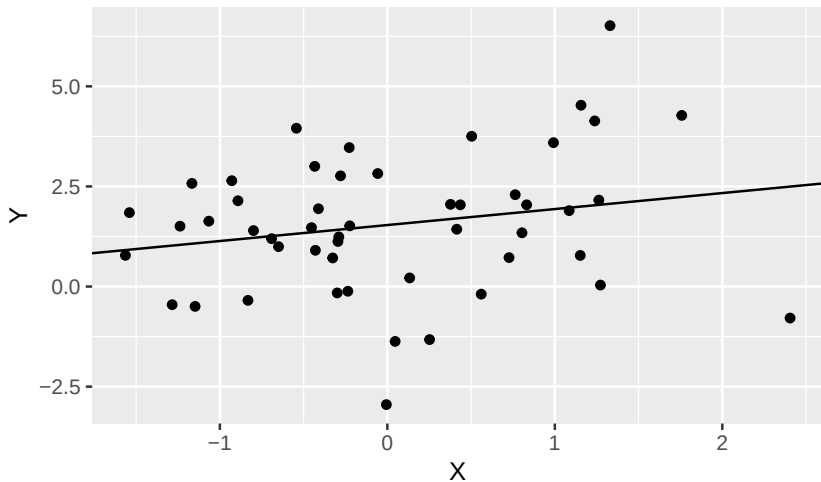
while the population median minimizes

$$\mathbb{E}(|Y - \mu|).$$

Visualizing the loss function

Let's take a random scatter plot of X and Y and view the loss function $L(\beta_0, \beta_1)$.

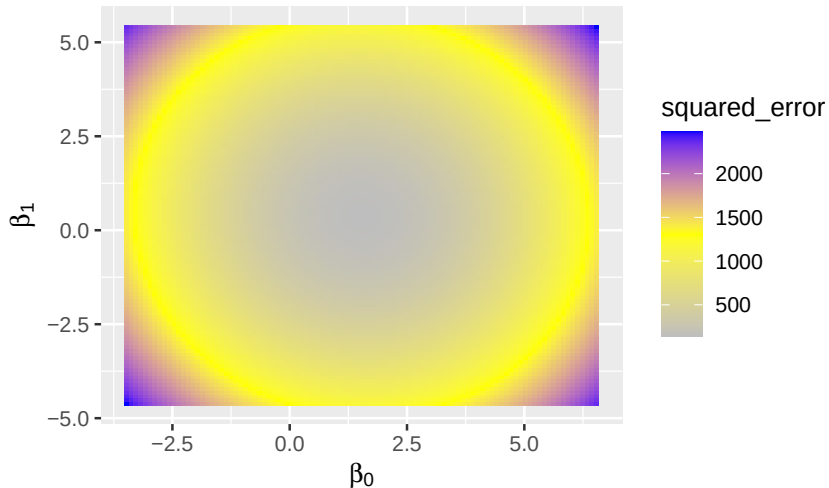
```
X = rnorm(50)
Y = 1.5 + 0.1 * X + rnorm(50) * 2
parameters.est = lm(Y ~ X)$coef
intercept = parameters.est[1]
slope = parameters.est[2]
```



- ▶ Let's plot the *loss* as a function of the parameters.
- ▶ Note that the *true* intercept is 1.5 while the *true* slope is 0.1.

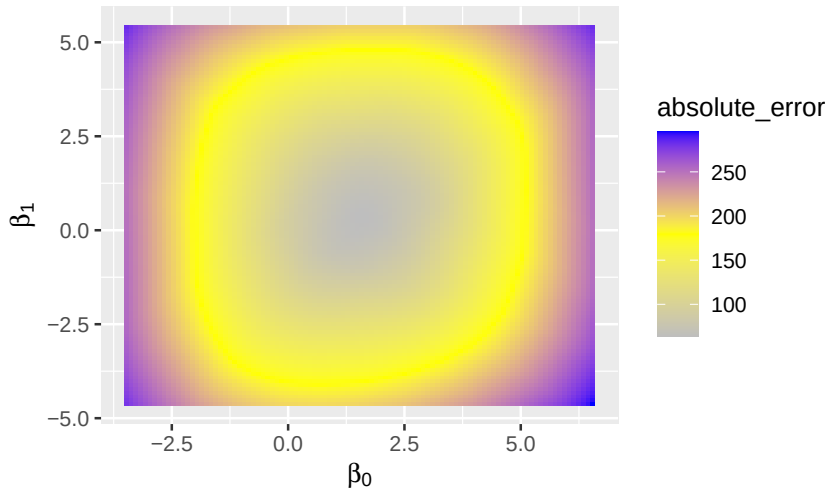
```
grid_intercept = seq(intercept - 5,
  intercept + 5, length = 100)
grid_slope = seq(slope - 5,
  slope + 5, length = 100)
loss_data = expand.grid(intercept_ = grid_intercept,
  slope_=grid_slope)

loss_data$squared_error = numeric(nrow(loss_data))
for (i in 1:nrow(loss_data)) {
  loss_data$squared_error[i] =
    sum((Y - X * loss_data$slope_[i] -
      loss_data$intercept_[i])^2)
}
```



Let's contrast this with the sum of absolute errors.

```
loss_data$absolute_error = numeric(nrow(loss_data))
for (i in 1:nrow(loss_data)) {
  loss_data$absolute_error[i] =
    sum(abs(Y - X * loss_data$slope_[i] -
           loss_data$intercept_[i]))
}
absolute_error_fig = (ggplot(loss_data,
  aes(intercept_, slope_,
    fill = absolute_error)) +
  geom_raster() +
  scale_fill_gradientn(colours = c("gray",
    "yellow", "blue")))
```

Geometry of least squares

- ▶ The following picture depicts the geometry involved in least squares regression.

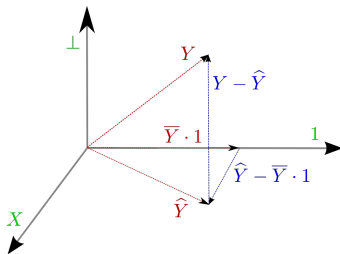


Figure 1: Source - Jonathan Taylor

- ▶ It requires some imagination but the picture should be thought as representing vectors in n -dimensional space, where n is the number of points in the scatter plot.
- ▶ In our height data, $n = 1375$. The bottom two axes should be thought of as 2-dimensional, while the axis marked “ $\perp$ ” should be thought of as $(n - 2)$ dimensional, or, 1373 in this case.

```
dim(Heights)
```

```
## [1] 1375    2
```

Estimation

Least squares estimators

- ▶ There are explicit formula for the least squares estimators, i.e. the minimizers of the error sum of squares.
- ▶ For the slope, $\hat{\beta}_1$, it can be shown that

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{\widehat{Cov}(X, Y)}{\widehat{Var}(X)}.$$

- ▶ Knowing the slope estimate, the intercept estimate can be found easily:

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \cdot \bar{X}.$$

Parameter Estimation

(cont.)

Properties of LSE

- ▶ $\sum_{i=1}^n e_i = 0$
- ▶ $\sum_{i=1}^n x_i e_i = 0$
- ▶ $\sum_{i=1}^n \hat{y}_i e_i = 0$

Distribution of $\hat{\beta}_1$

Distribution of $\hat{\beta}_0$

Important lengths

- We can describe an observation as

$$\underbrace{y_i}_{\text{Observed}} = \underbrace{\hat{y}_i}_{\text{Fit}} + \underbrace{(y_i - \hat{y}_i)}_{\text{Deviation from fit}}.$$

- Subtract $\bar{y}$ from both sides

$$\underbrace{y_i - \bar{y}}_{\text{Deviation from mean}} = \underbrace{\hat{y}_i - \bar{y}}_{\text{Deviation due to fit}} + \underbrace{(y_i - \hat{y}_i)}_{\text{Residual}}.$$

- The (squared) lengths of the vectors $(\mathbf{Y} - \hat{\mathbf{Y}})$, $(\bar{\mathbf{Y}} - \hat{\mathbf{Y}})$, $(\mathbf{Y} - \bar{\mathbf{Y}})$ are important quantities in what follows.

$$SSE = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 = \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2$$

$$SSR = \sum_{i=1}^n (\bar{Y} - \hat{Y}_i)^2 = \sum_{i=1}^n (\bar{Y} - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2$$

$$SST = \sum_{i=1}^n (Y_i - \bar{Y})^2 = SSE + SSR$$

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST} = \widehat{Cor}(\mathbf{X}, \mathbf{Y})^2.$$

Coefficient of determination

An important summary of the fit is the ratio

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

which measures *how much variability in Y* is explained by *X*.

Estimate of σ^2

- ▶ There is one final quantity needed to estimate all of our parameters in our (statistical) model.
- ▶ This is σ^2 , the variance of the random variation within each slice (the regression model assumes this variance is constant within each slice).
- ▶ The estimate most commonly used is

$$\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2 = \frac{\text{SSE}}{n-2} = \text{MSE}$$

- ▶ Above, note the practice of replacing the quantity $SSE(\hat{\beta}_0, \hat{\beta}_1)$, i.e. the minimum of this function, with just SSE .
- ▶ The term MSE above refers to mean squared error: a sum of squares divided by what we call its *degrees of freedom*.
 - ▶ The degrees of freedom of SSE , the *error sum of squares* is therefore $n - 2$.
 - ▶ Remember this $n - 2$ corresponded to $\perp$ in the picture above.

- ▶ Using some statistical calculations that we will not dwell on, if our simple linear regression model is correct, then we can see that

$$\frac{\hat{\sigma}^2}{\sigma^2} \sim \frac{\chi_{n-2}^2}{n-2}$$

where the right hand side denotes a *chi-squared* distribution with $n - 2$ degrees of freedom.

- ▶ (Note: our estimate of σ^2 is *not* the maximum likelihood estimate.)

Example wages vs. education

- ▶ In this example, we'll look at the output of *lm* for the wage data and verify that some of the equations we present for the least squares solutions agree with the output.
- ▶ The data was compiled from a study in econometrics [Learning about Heterogeneity in Returns to Schooling](#).

```
fname = './wage.csv'  
wages = read.table(fname, sep=',', header=TRUE)  
print(head(wages))
```

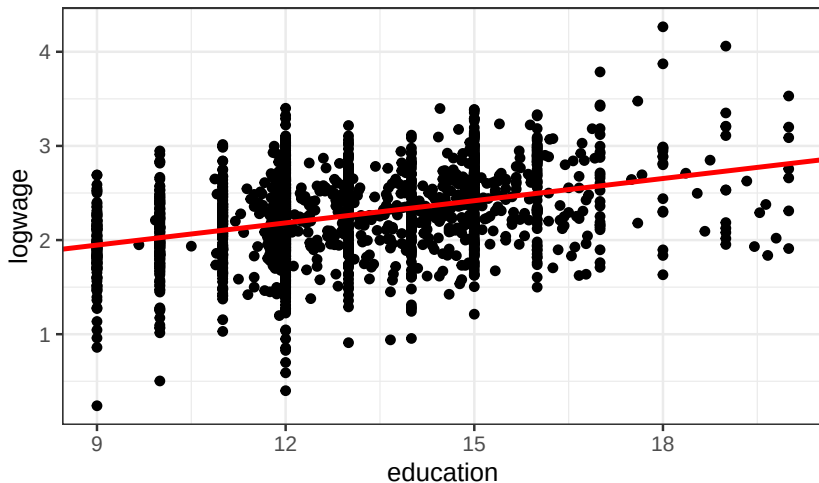
```
##   Person_id education  logwage  
## 1          1  16.75000 2.845000  
## 2          2  15.00000 2.446667  
## 3          3  10.00000 1.560000  
## 4          4  12.66667 2.099167  
## 5          5  15.00000 2.490000  
## 6          6  15.00000 2.330833
```

- Let's fit the linear regression model.

```
wages.lm = lm(logwage ~ education,
  data = wages)
print(wages.lm)
```

```
##
## Call:
## lm(formula = logwage ~ education, data = wages)
##
## Coefficients:
## (Intercept)      education
##      1.2392         0.0786
```

- ▶ As in the mother-daughter data, we might want to plot the data and add the regression line.



- Compute the least squares estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ using the formula

```
beta.1.hat = cov(wages$education,
  wages$logwage) / var(wages$education)
beta.0.hat = mean(wages$logwage) -
  beta.1.hat * mean(wages$education)
```

- Compare the above with the lm output

```
print(c(beta.0.hat, beta.1.hat))
```

```
## [1] 1.23919433 0.07859951
```

```
print(coef(wages.lm))
```

```
## (Intercept)    education
```

```
## 1.23919433 0.07859951
```

- Compute $\hat{\sigma}^2$ using the formula

```
sigma.hat = sqrt(sum(resid(wages.lm)^2) /  
  wages.lm$df.resid)  
c(sigma.hat, sqrt(sum((wages$logwage -  
  predict(wages.lm))^2) / wages.lm$df.resid))
```

```
## [1] 0.4037828 0.4037828
```

- The summary from *R* also contains this estimate of σ : (Residual standard error)

```
summary(wages.lm)
```

```
##
## Call:
## lm(formula = logwage ~ education, data = wages)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.78239 -0.25265  0.01636  0.27965  1.61101
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.239194    0.054974   22.54  <2e-16 ***
## education    0.078600    0.004262   18.44  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4038 on 2176 degrees of freedom
## Multiple R-squared:  0.1351, Adjusted R-squared:  0.1347
## F-statistic: 340 on 1 and 2176 DF,  p-value: < 2.2e-16
```

References for this lecture

- ▶ Based on the lecture notes of [Pratheepa Jeganathan](#)
- ▶ Based on the lecture notes of [Jonathan Taylor](#) .
- ▶ Lecture notes of [Stats 205](#)